



Test-Time Failure Prediction for Spacecraft Pose Estimation Under Domain Shift

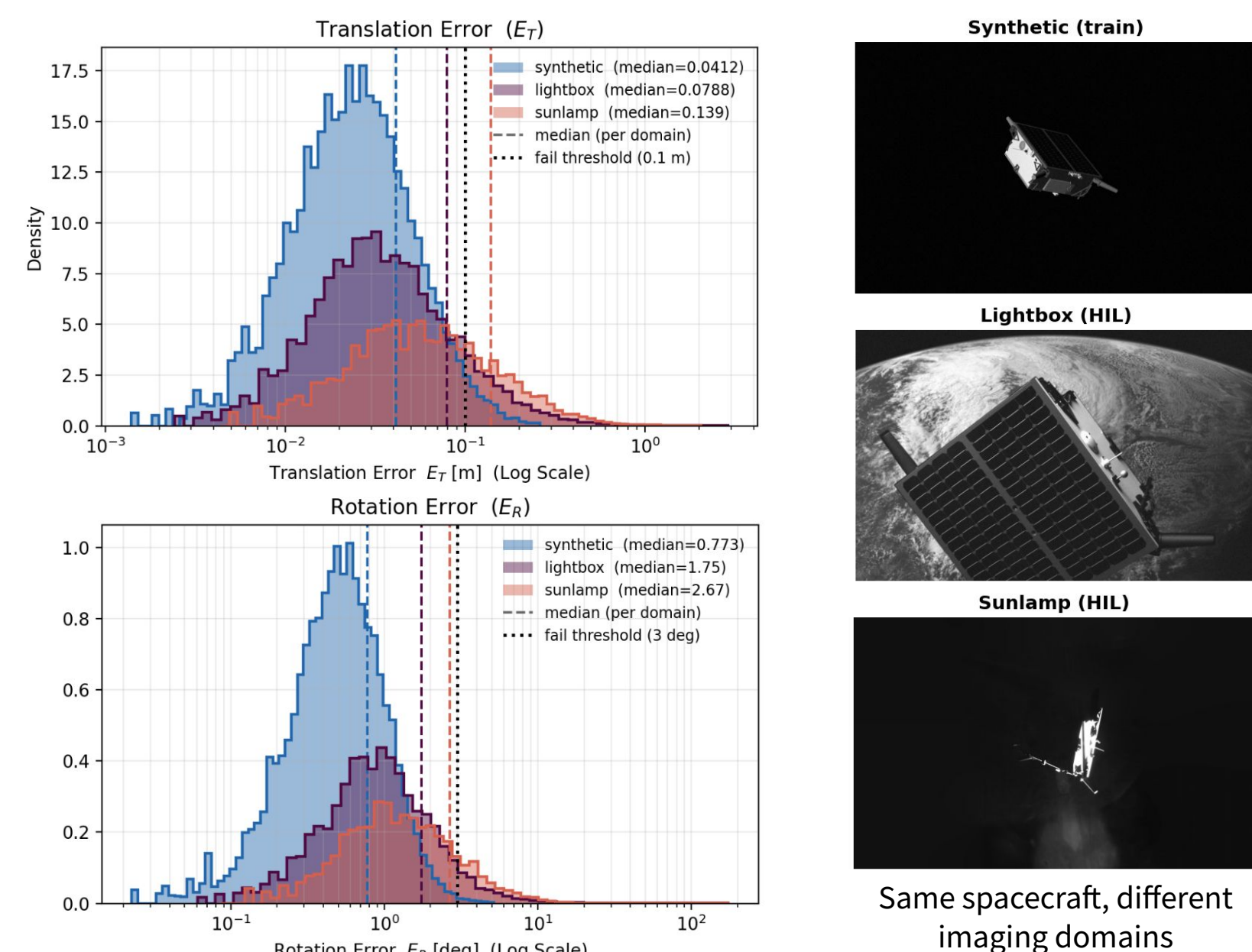
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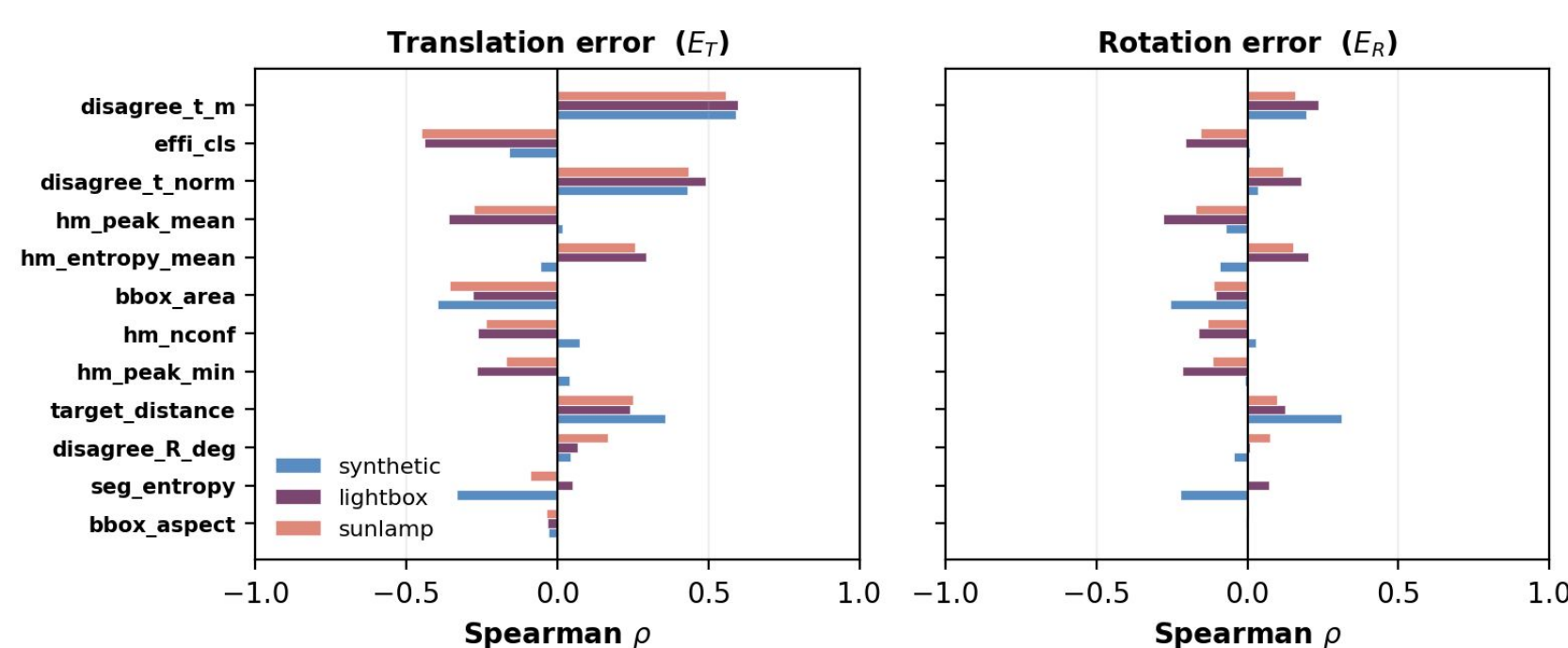
Overview

- Motivation:** No reliable uncertainty signal exists for spacecraft pose estimators despite being critical for safe autonomous proximity operations (docking, debris removal, etc.)
 - Models train on synthetic data but deploy in harsh real-world lighting → creates a **domain gap** that **degrades reliability**
 - Navigation filters use per-measurement uncertainty to reject bad poses before they corrupt the state estimate
- We build a **lightweight binary failure predictor** for SPNv2 (a SOTA pose estimator) that **generalizes across** the synthetic-to-real **domain gap** using only **12 internal network features**, no model changes, and **only using synthetic labels**
- Our **direct-transfer logistic regression model** achieves **AUC 0.84** (lightbox) and **0.80** (sunlamp) for **translation error failure prediction**, nearly matching an in-domain oracle



Data & Features

- Data:** **SPEED+ dataset** – 11,994 synthetic validation images (disjoint from SPNv2's training split) and 9,524 **HIL test images** across two illumination domains (**lightbox**: diffuse, **sunlamp**: direct). **Synthetic poses provide the only training labels**; HIL poses are reserved for evaluation
- Feature Selection:** 12 internal SPNv2 signals extracted from a single forward pass on the image; largely domain-invariant

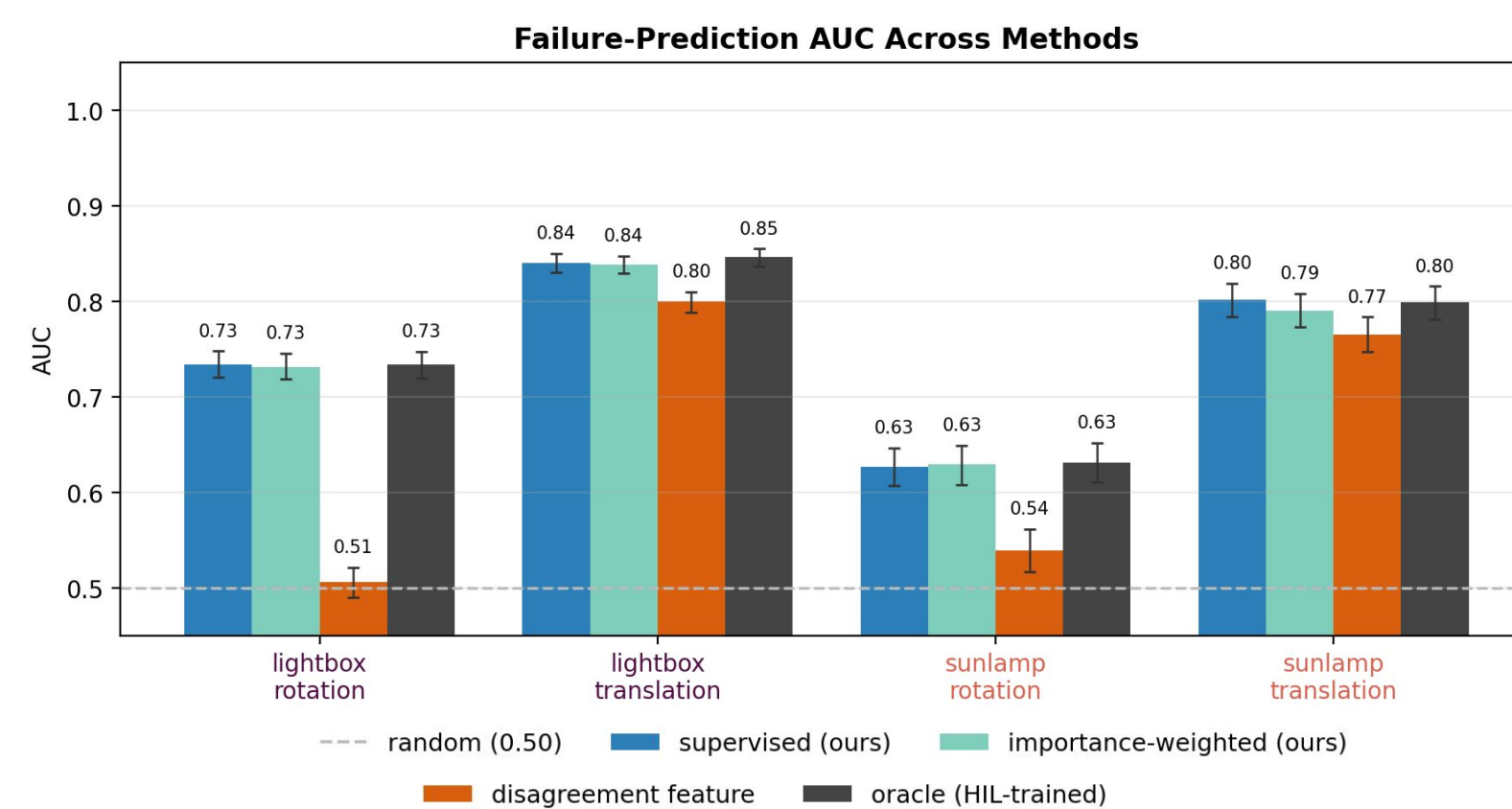
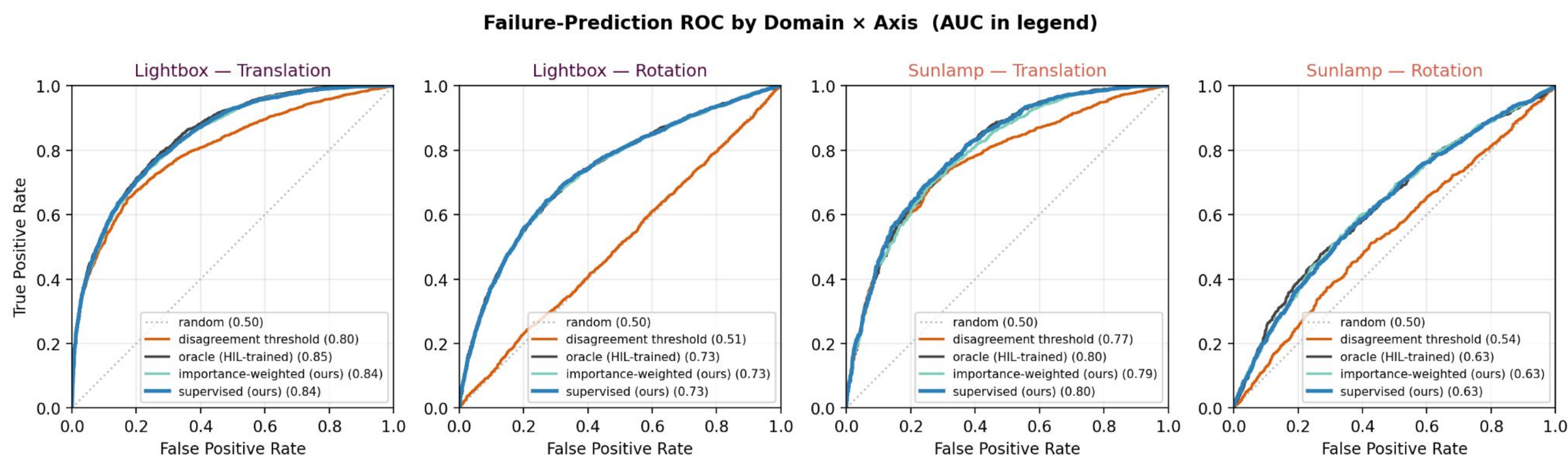


Models

Random Predictor	Flags failures at the synthetic failure rate (11% translation, 2% rotation)
Single Feature	Thresholds SPNv2's internal head disagreement at the F1-optimal cutoff on synthetic data → strong single-feature signal our model should outperform.
Supervised Failure Predictor	ℓ_2 -regularized, class-weighted logistic regression over 12 SPNv2 internal features, trained on synthetic labels and applied unchanged to HIL. Each sample weighted by inverse class frequency. $y = 1[e > \tau], \quad \tau_T = 0.10 \text{ m}, \quad \tau_R = 3.0^\circ$ $\mathcal{L} = - \sum_i \alpha_{y_i} [y_i \log \hat{p}_i + (1-y_i) \log(1-\hat{p}_i)] + \frac{\lambda}{2} \ \theta\ _2^2 \quad \alpha_c \propto 1/n_c$
Importance-Weighted Predictor	Same as above, but reweights synthetic training examples to correct for domain shift . Weights are estimated via a logistic regression domain classifier trained on synthetic and HIL features (no failure labels). By Bayes' rule, the density ratio reduces to the $w(x) \propto \frac{d(x)}{1-d(x)}$ ds, so synthetic examples whose features look HIL-like receive higher weight $w(x)$.
Oracle Upper Bound	Logistic regression trained directly on HIL labels (5-fold CV), isolating the cost of the domain gap.
Unsupervised Anomaly	Four unsupervised models (Mahalanobis, GMM, one-class SVM, isolation forest) fit on synthetic features ; anomaly scores weakly correlate with HIL pose error but are substantially less effective than supervised prediction.

*One model per error axis per test domain. Key → baseline; method; upper bound; other

Results



Operating-Point Metrics for Direct Supervised Predictor

	Threshold	Fail Rate	Precision	Recall	F1
E_T	0.73	0.41	0.75	0.63	0.69
E_R	0.91	0.27	0.55	0.47	0.51
E_T	0.73	0.63	0.85	0.62	0.72
E_R	0.91	0.45	0.59	0.42	0.49

Failure decision threshold selected per error axis (E_T [translation, m], E_R [rotation, degrees]) to maximize F1 on a synthetic validation split, then frozen and applied to **lightbox** and **sunlamp** HIL domains.

Discussion

- A single SPNv2 forward pass carries a **label-free failure signal** that allows us to flag translation errors well (**AUC 0.84** lightbox, **0.80** sunlamp), which beats both our baselines
- Importance weighting gave no improvement** over direct transfer (within ± 0.01 AUC), which indicates the synthetic feature to failure mapping already transfers to HIL, and our synthetic-trained model **nearly matches the in-domain oracle** (gap ≤ 0.006 AUC)
- Rotation is the weak axis**. No feature carries a strong rotation signal, and synthetic rotation failures are rare (**only 2%**), which starves the classifier of positive examples despite class balancing
- Sunlamp is the hardest domain** to predict due to the larger domain gap, with domain classifier **AUC 0.91** vs **0.82** for lightbox
- Overfitting risk is low** (12 features, ~12k examples, ℓ_2 regularization), and the oracle confirms that our models generalize to HIL

Future Work

- Closing the rotation gap** is the priority. No single feature correlates strongly with rotation error, so we would need to **expand the feature set** to include rotation and illumination-sensitive signals
- We **classify against a fixed error threshold**, which throws away information. **Regressing directly onto continuous, physical error** (predicting E_T/E_R in meters and degrees) would give filters a better calibrated magnitude to weigh their estimates on
- Finally, we would like to **integrate our method into a downstream adaptive UKF** that dynamically scales the measurement-noise covariance to outright reject or down-weight bad pose estimates before they corrupt the state estimate

References

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